

WASP-52b

R-1886

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-52_190813 · created 2026-08-03T12:25:39+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458708.8092 +/- 0.0049 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.039 +/- 0.061
Transit depth (Rp/Rs)^2	0.15 +/- 0.48 [%]
Orbital Inclination (inc)	81.83 +/- 2.8
Airmass coefficient 1 (a1)	1.5 +/- 0.98
Airmass coefficient 2 (a2)	-0.44 +/- 0.46
Scatter in the residuals of the lightcurve fit is	55.35 %
Best Comparison Star	#1 - [115, 151]
Optimal Aperture	1.34
Optimal Annulus	4.25
Transit Duration (day)	0.026 +/- 0.028

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.039 ± 0.061 vs 0.1646 (deviation 2.05σ)
Duration fit vs published	0.026 d vs 0.0758 d (deviation 1.77σ)
Mid-time O–C	-11.3 min
Depth SNR	0.6
Residual scatter	55.35 %

Light curve

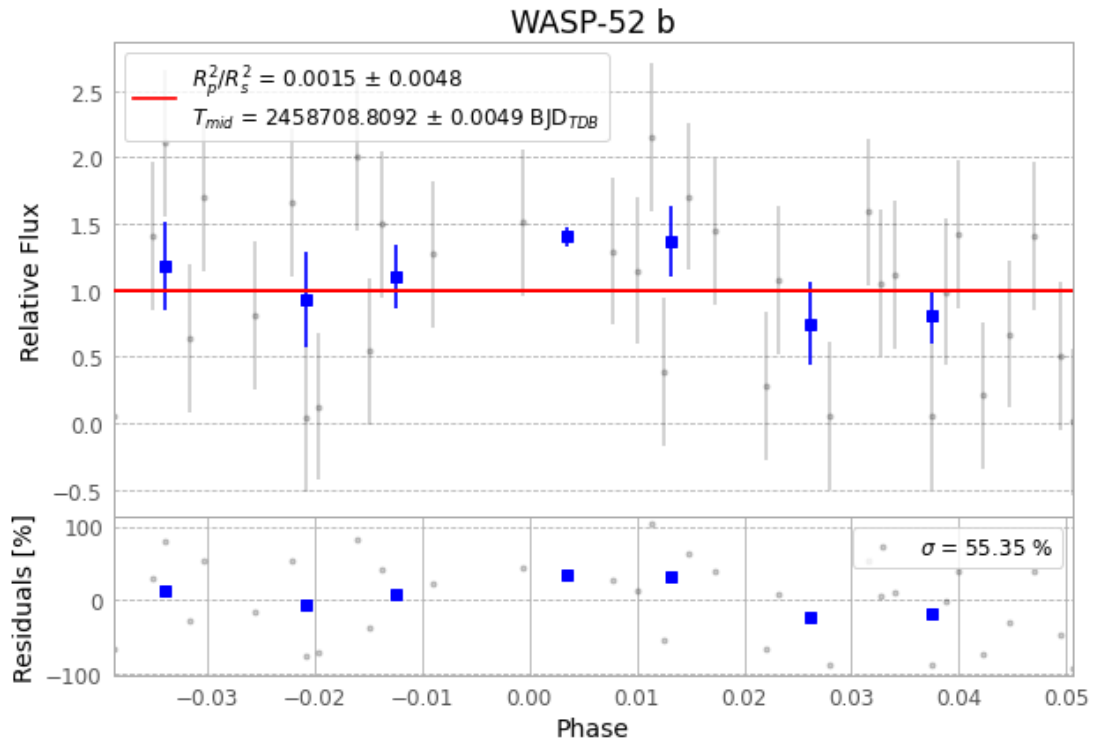


Figure 1 — FinalLightCurve_WASP-52 b_2019-08-12.png (SHA-256 1d63471500c0689e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [462, 127], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 5d902c1ae3dd3f1b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

77 FITS frames (51 MB), WASP-52190813053912.FITS → WASP-52190813092712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-190813.log (b19088de4e03...), FinalLightCurve_WASP-52 b_2019-08-12.png (1d63471500c0...), FinalLightCurve_WASP-52 b_2019-08-12.csv (7bf05bb52b86...)

Compute resources

Wall time 155.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 12:25Z.